Operating System

Instructor: Dr. Qiyu Liu

Southwest University

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About Myself

Qiyu Liu

Ph.D. in Computer Science

Qiyu Liu

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Research Interest:

- Database System: Indexing, Query Processing, Query Optimization...
- AI System: LLM Inference Acceleration, Distributed Training...
- HPC: SIMD, CUDA, New Hardware...

About the Course

Schedule:

- Lecture: 19:20-21:55 Every Thursday Room 16-0305 (Week 1-16)
- Experiments: 10:00-11:40 Every Tuesday Room 25-0702 (Week 3-14)
- Tutorial: TBD if possible
- Office Hour: (Mostly) 10:00-12:00 Friday

Kind Reminder:

Email me if you want to make an appointment in other time slot :)

About the Course (Cont.)

Grading System:

- Attendance + Homework: 20%
- Project: 15% (in total 5 projects)
- Final Exam: 60%

Very Important:

- Please request leave in advance if you cannot attend the lecture.
- **No cheating and no bargaining!**
- I ensure that u can learn a lot if you attend all my classes :)

About the Course (Cont.)

Course Resources: https://github.com/qyliu-hkust/SimpleOS
Textbook: Operating System Concepts (10-th Edition)
References:

- OSTEP: https://pages.cs.wisc.edu/~remzi/OSTEP/
- The OS textbook from Prof. Haibo Chen. (Chinese version)
- Algorithmica: https://github.com/algorithmica-org/algorithmica

Objectives

- Describe the general **organization** of a computer system and the role of **interrupts**
- Describe the components in a **modern multiprocessor** computer system
- Illustrate the transition from **user mode** to **kernel mode**
- Discuss how operating systems are used in various computing environments
- Provide examples of free and open-source operating systems

What is an OS?

- Windows/Linux/Android is an OS?
- I am an OS?
- My cats are OS?

What is an OS? (Cont.)

- A program that acts as an **intermediary** between a user of a computer and the computer hardware
- Operating system goals:
  1. Execute user programs and make solving user problems easier
  2. Make the computer system convenient to use
  3. Use the computer hardware in an efficient manner

Rethinking "Hello World!"

- Use stdio.h: printf("Hello World!");
- Use unistd.h: write(1, "Hello World!", 12); (What?)
- **Even Horrible!**

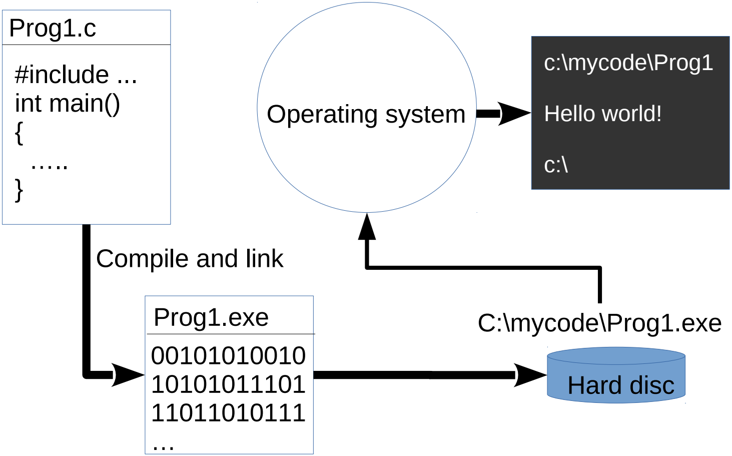
```
section .data
    msg db "Hello, World!", 10   ; 10 是换行符 '\n'
    len equ $ - msg              ; 计算字符串长度

section .text
    global _start

_start:
    ; syscall write(STDOUT_FILENO, msg, len)
    mov rax, 1        ; syscall number for sys_write (1)
    mov rdi, 1        ; file descriptor 1 (stdout)
    mov rsi, msg      ; 指向字符串的指针
    mov rdx, len      ; 字符串长度
    syscall           ; 调用内核

    ; syscall exit(0)
    mov rax, 60       ; syscall number for sys_exit (60)
    xor rdi, rdi      ; exit code 0
    syscall           ; 调用内核
```

Rethinking "Hello World!" (Cont.)

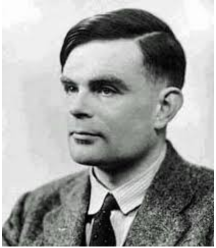


Computer System

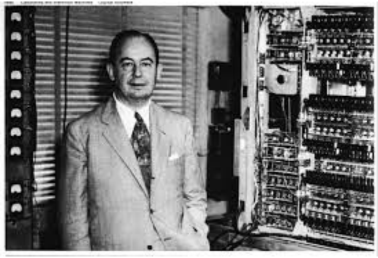
Computer system includes:

- Hardware (CPU, memory, I/O devices) - Provides basic computing resources
- Operating system - Controls and coordinates use of hardware among various applications and users
- Application programs - Define the ways in which the system resources are used to solve the computing problems of the users
- Users - People, other machines

Von Neumann Architecture

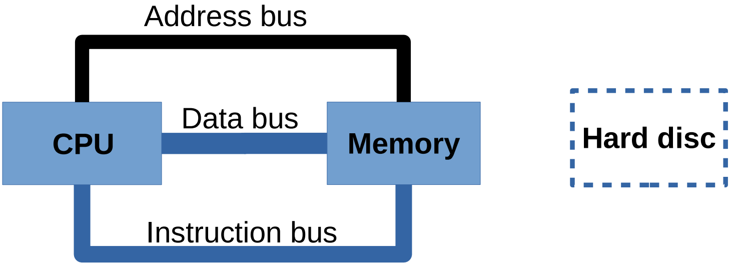


Alan Turing
(1912-1954)

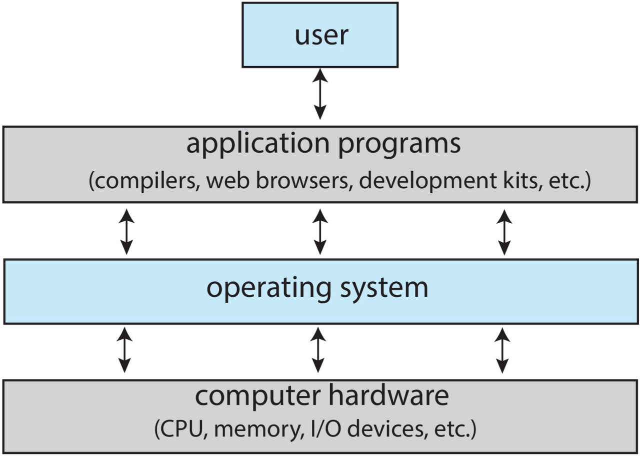


John Von Neumann
(1903-1957)

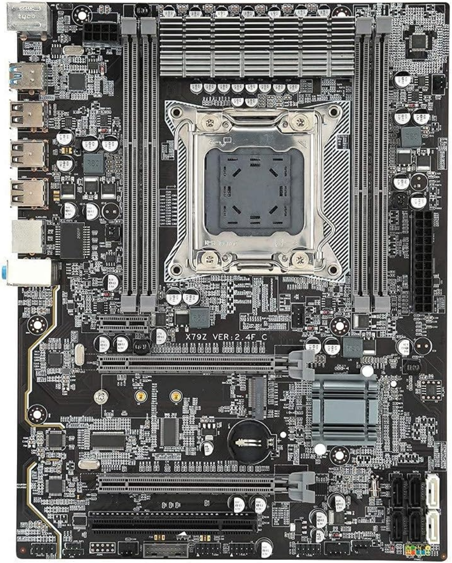
Von Neumann Architecture (Cont.)



Computer System (Cont.)



A Motherboard



Do you know these hardwares?

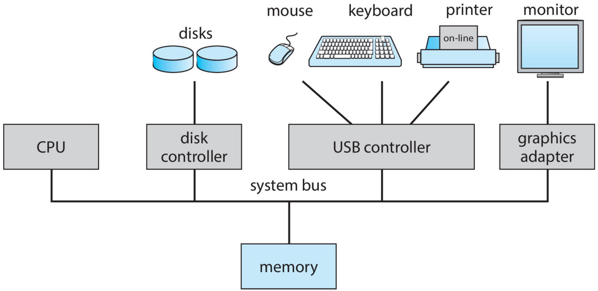






Computer System Operation

- One or more CPUs, device controllers connect through common bus providing access to shared memory
- Concurrent execution of CPUs and devices competing for memory cycles



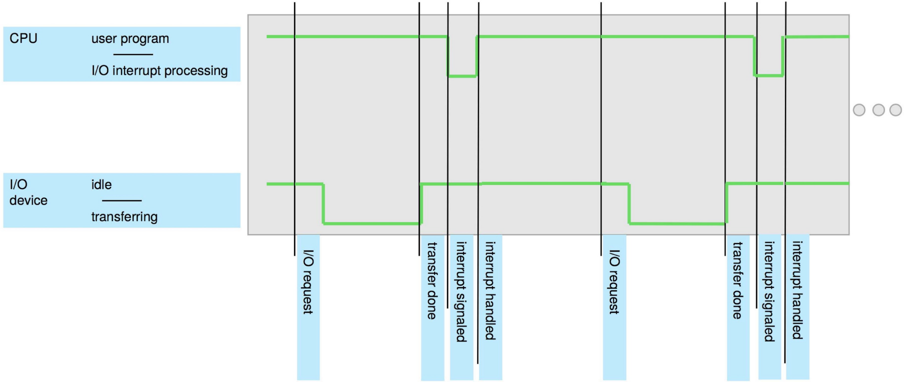
Computer System Operation (Cont.)

- I/O devices and the CPU can execute concurrently
- Each device controller is in charge of a particular device type
- Each device controller has a local buffer
- Each device controller type has an operating system device driver to manage it
- CPU moves data from/to main memory to/from local buffers
- I/O is from the device to local buffer of controller
- Device controller informs CPU that it has finished its operation by causing an **interrupt**

Why we need interrupts?

- Interrupt transfers control to the interrupt service routine generally, through the **interrupt vector**, which contains the addresses of all the service routines
- Interrupt architecture must save the address of the interrupted instruction
- A trap or exception is a software-generated interrupt caused either by an error or a user request
- An operating system is interrupt driven

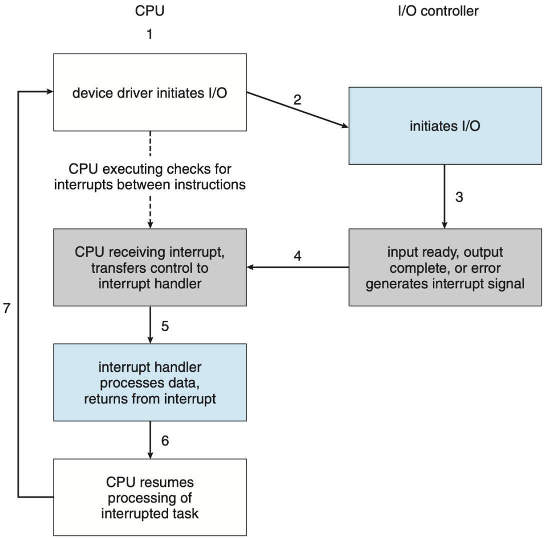
Why we need interrupts? (Cont.)



Interrupt Handling

- The operating system preserves the state of the CPU by storing the registers and the program counter
- Determines which type of interrupt has occurred
- Separate segments of code determine what action should be taken for each type of interrupt

An I/O Example



OS Objective (Again)

Q & A